

SAT Hard Question Set 3

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Phase 2 · Hard Question Practice

Overview

1 Hard Question Set 3

Question 1

A research manager selected 2 random samples of ovens of a certain type to estimate the average amount of time this type of oven takes to preheat to 325°F . The research manager recorded the amount of time, in minutes, each oven takes to preheat to 325°F .

Based on the first sample, the manager estimated an average of 14.2 minutes, with a margin of error of 1 minute. Based on the second sample, the estimate was 14.4 minutes, with a margin of error of 2.3 minutes.

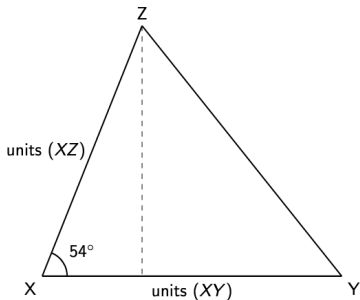
Assuming the margins of error were calculated the same way, which of the following best explains why the first sample had a smaller margin of error?

- A. The first sample contained fewer ovens than the second sample.
- B. The first sample contained more ovens than the second sample.
- C. The first sample took less time on average to preheat.
- D. The first sample took more time on average to preheat.

Answer 1

Correct Answer: B

Question 2



In triangle XYZ , $\angle X = 54^\circ$, $XY = 26$ units, $XZ = 19$ units. What is the area, in square units, of triangle XYZ ?

- A. 247
- B. 494
- C. $247 \sin 54^\circ$
- D. $494 \sin 54^\circ$

Answer 2

Correct Answer: C

Question 3

In the figure, $\triangle ABC$ is inscribed in a semicircle with diameter $AC = 10$. Point B lies on the semicircle. If $AB = 6$ and $BC = x$, find x . Give your answer in simplest radical form.

- A. $\sqrt{36}$
- B. 8
- C. $\sqrt{44}$
- D. $\sqrt{64}$

Answer 3

Correct Answer: B

Question 4

A parabola is given by

$$y = x^2 - 4x + 5,$$

and a circle by

$$x^2 + y^2 = 25.$$

The parabola and circle intersect at points P and Q . Find the distance between P and Q .

Answer 4

Final Answer: 3.88

Question 5

Solve the equation:

$$\sqrt{6r} = \sqrt{r^2 - 16}$$

What is the product of all solutions to the above equation?

- A. 4
- B. 6
- C. 8
- D. 12

Answer 5

Correct Answer: C

Thank You!



We appreciate your time and focus.

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